

Cooling, maintenance, and the energy it saves.

Three questions asked while preparing the UNIDO presentation, with sourced answers. Full arithmetic and source links: reference/tco-research.md.

Q1 How much of the world's electricity do all air conditioners use?

"Hoeveel energie verbruiken alle airco's ter wereld samen van de totale opgewekte stroom?"

About 10% of all electricity generated worldwide goes to air conditioning and space cooling. [IEA](#)

- **~2,000 TWh in 2016**, about 10% of global electricity consumption (ACs + fans).
[IEA · The Future of Cooling, 2018](#)
- **~2,400 TWh in 2022**, still just under 10% of global final electricity demand, and rising in absolute terms. [IEA tracking, 2022](#)
- Projection: cooling demand roughly **triples to ~6,200 TWh by 2050** as the stock grows from ~2 billion to ~5.6 billion units. [IEA](#)

Deck phrasing: "Space cooling already uses about 10% of the world's electricity (IEA)." A neglected unit burns up to 15% more to deliver the same cooling (US DOE), so maintenance is a direct lever on that 10% block.

Q2 How much electricity does maintenance save, per year and over 20 years?

"Hoeveel elektriciteit spaar je per jaar over een 20-jaar periode door onderhoud?"

ONE SPLIT AC (12K BTU, URUGUAY MODEL)	SAVED
Normal use, maintained	900 kWh/yr Ecool.uy · UTE 2026
Neglect penalty avoided (+15%, conservative)	~135 kWh ≈ US\$35 per year US DOE composite
Over 20 years	~2,700 kWh ≈ US\$695 (nominal, UTE \$0.257/kWh)
At scale: 100 units	~13,500 kWh/yr · ~270 MWh ≈ US\$69,000 over 20 yrs

- Over 20 years a neglected AC wastes the equivalent of **~3 full years of its own normal consumption**.
- +15% is conservative: dirty filter alone costs 5–15%, fouled coils up to 30%, wrong refrigerant charge –15% efficiency; most field units run out of spec. [US DOE](#) · [ENERGY STAR](#) · [Purdue](#) A badly neglected unit sits at +20–30% → 3,600–5,400 kWh/20 yrs.
- Boilers are the gas twin: **up to +8–10% fuel** when unserviced (≈US\$160/yr at NL/DE prices).
[trade sources](#)
- Maintenance fees are already repaid by the avoided third unit and avoided emergency repairs, so this energy saving is **pure profit on top**.

Q3 What is the total consumption of ACs worldwide, in absolute terms?

"Wat zou dan het totale verbruik van airco's wereldwijd zijn?"

Roughly 2,400 TWh per year, the world's air-conditioning electricity bill. IEA 2022

- $\approx$ **60% of the entire US grid's** annual consumption, running just to cool air.
- $\approx$ **~200× Uruguay's total electricity consumption** (~11–12 TWh/yr): the world's ACs use two hundred Uruguays.
- Maintenance kicker (our estimate, flagged as such): if even half the world's stock runs 10–15% over from neglect, that is **~100–200 TWh wasted per year, roughly the Netherlands' entire annual electricity consumption**, burned by dirty filters, fouled coils and bad charge.
- Maintenance is a country-sized energy-efficiency lever: squarely the UNIDO / SDG 7 agenda.